Project Design Phase

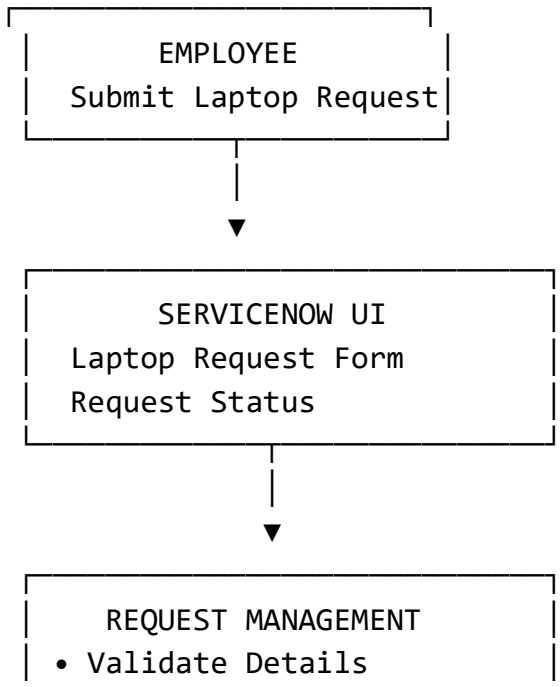
Solution Architecture

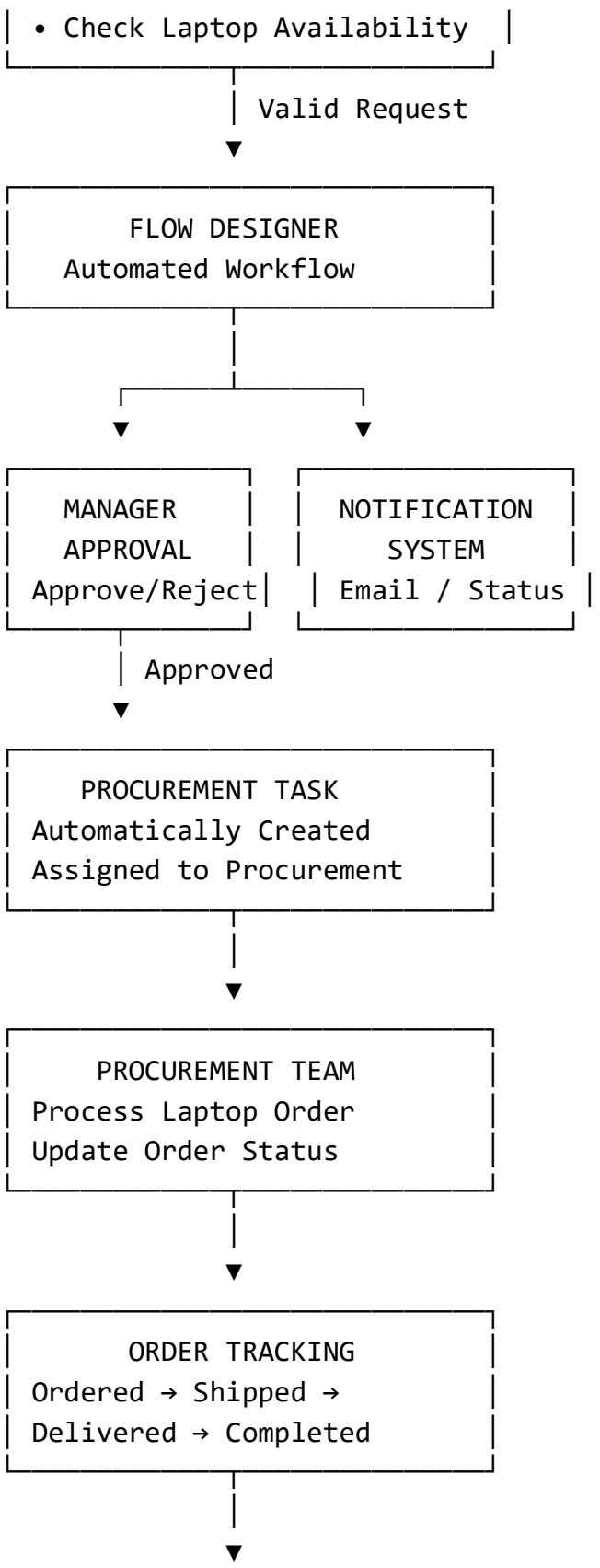
Date	03 October 2026
Team ID	SWTID-2026-5094
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Solution Architecture

The solution architecture describes how **ServiceNow** components work together to automate the standard laptop procurement process. The architecture connects the employee request, validation, manager approval, procurement task, notifications, and order tracking into one automated workflow.

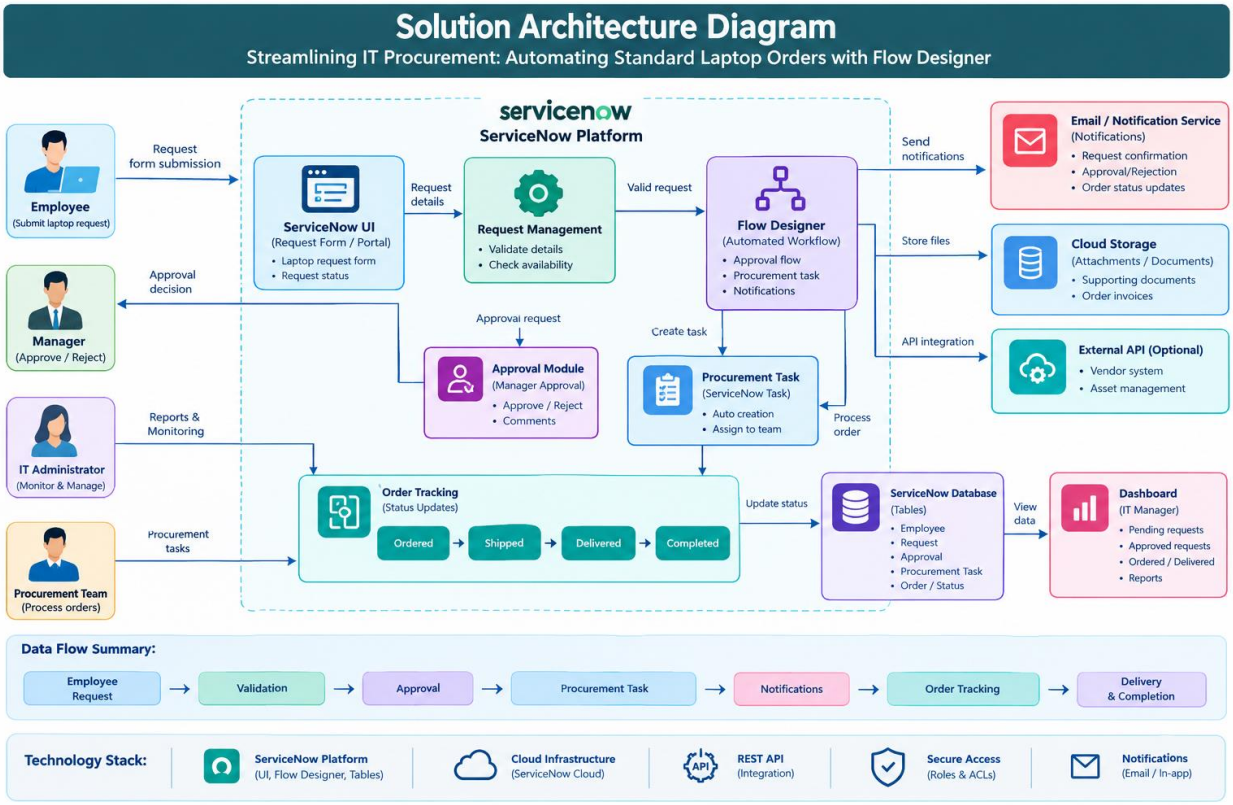
Figure 1: Solution Architecture and Data Flow




• Check Laptop Availability
Valid Request
FLOW DESIGNER
Automated Workflow
MANAGER APPROVAL
Approve/Reject
NOTIFICATION SYSTEM
Email / Status
Approved
PROCUREMENT TASK
Automatically Created
Assigned to Procurement
PROCUREMENT TEAM
Process Laptop Order
Update Order Status
ORDER TRACKING
Ordered → Shipped →
Delivered → Completed

```
┌──────────────────────────┐
│        EMPLOYEE          │
│ Status & Completion      │
│ Notification             │
└──────────────────────────┘
```

```
┌──────────────────────────────────────────┐
│        SERVICENOW DATA STORAGE           │
│ Employee | Request | Approval            │
│ Procurement Task | Order | Status        │
└──────────────────────────────────────────┘
```



Architecture Components

Component	Purpose
Employee Interface	Allows employees to submit and track laptop requests.
Request Management	Collects request information and validates mandatory details.

Flow Designer	Automates the complete procurement workflow.
Approval Module	Allows managers to approve or reject requests.
Procurement Task	Automatically creates and assigns procurement work.
Notification System	Sends request, approval and order-status notifications.
Order Tracking	Tracks the laptop from order placement to delivery.
ServiceNow Tables	Store employee, request, approval, task and order information.
Dashboard	Allows IT managers to monitor procurement progress.

Data Flow

Employee Request → Validation → Flow Designer → Manager Approval → Procurement Task → Order Processing → Status Updates → Delivery → Completion

Key Benefits of the Architecture

- Reduces manual procurement activities.
- Automates approval and task creation.
- Provides clear request and order tracking.
- Improves communication through automatic notifications.
- Provides role-based access for employees, managers, IT administrators and procurement teams.
- Creates a centralized procurement workflow on ServiceNow.